

LaTeX Test Document

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1 Introduction

This is a simple test document to verify the L^AT_EX compilation pipeline.

2 Mathematics

Here is a fundamental equation in ML:

$$z = \frac{x - \mu}{\sigma}$$

3 Python Code

```
1 def calculate_z_score(x, mu, sigma):  
2     """Calculates the standard score."""  
3     return (x - mu) / sigma
```

4 Callout Boxes

ML CONNECTION:

The z -score is used for standardization in preprocessing pipelines to ensure features have zero mean and unit variance.

5 Table Test

Feature	Mean	Std Dev
Age	35.5	12.1
Income	\$65,000	\$15,000